

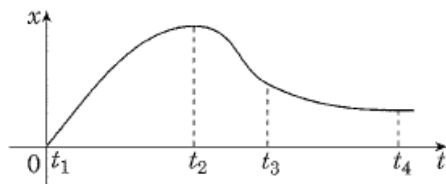
# APPLICATIONS OF CALCULUS TO THE PHYSICAL WORLD

## Examination questions

- 1 The displacement  $x$  metres from the origin at time  $t$  seconds, of a particle travelling in a straight line is given by the formula  $x = t^3 - 6t^2 + 1$ .
  - (a) Find the times when the particle is at rest.
  - (b) Find the acceleration of the particle at time  $t$  seconds.
- 2 A particle moves in a straight line such that its distance  $x$ , in metres, from a fixed point  $O$  is given by  $x = 2 - 4\sin 2t$  where  $t$  is the time, measured in seconds, commencing at  $t = 0$ .
  - (i) What is the particle's initial position?
  - (ii) At what time, and where, does the particle first come to rest?
  - (iii) At what time does the particle next come to rest?
  - (iv) What is the acceleration of the particle when  $t = \frac{\pi}{12}$  seconds?
- 3 After time  $t$  seconds, the displacement  $x$  metres from a fixed point  $O$  of a particle moving in a straight line is given by  $x = f(t) = 3 - \cos 2t$ ,  $0 \leq t \leq 2\pi$ .
  - (a) Sketch the graph of  $x = f(t)$ .
  - (b) Determine the times at which the particle is at rest and find the position of the particle at each of these times.
  - (c) Find the time when the particle reaches its maximum velocity for the first time.
- 4 The displacement  $x$  metres from the origin at time  $t$  seconds of a particle travelling in a straight line is given by  $x = t^3 - 12t$ , where  $t \geq 0$ .
  - (i) Find the velocity at time  $t$  seconds.
  - (ii) Calculate the velocity of the particle when  $t = 3$ .
  - (iii) Find the time when the particle is stationary.
- 5 A particle moves on a straight line. Its distance  $x$  metres from a fixed point  $O$  on the line is given by  $x = 5 - t + 4\ln(t+2)$ , where the time  $t$  is measured in seconds.
  - (i) What is the position of the particle when  $t = 0$ .
  - (ii) Find both the velocity and acceleration of the particle at time  $t$  seconds.
  - (iii) Find the time  $t$  when the particle is stationary.
- 6 Two particles  $A$  and  $B$  commence to move along the  $x$ -axis at time  $t = 0$ . The position of  $A$  at time  $t$  is given by  $x_A = \frac{1}{2}t + 1$  and the position of  $B$  is given by  $x_B = \ln(t+1)$ . The distances travelled by  $A$  and  $B$  are measured in centimetres.
  - (i) Find expressions for the velocities of the two particles.
  - (ii) On the same set of axes draw the graphs of the velocities for the particles  $A$  and  $B$  and show that there is only one occasion when the particles have the same velocity.
  - (iii) Show that when the particles have the same velocity, particle  $B$  has travelled  $\ln 2$  centimetres from its start position.
  - (iv) Show that the two particles will never meet.
- 7 A moving particle has an acceleration  $a$  metres per second per second at time  $t$  seconds ( $t \geq 0$ ). The acceleration of the particle is given by  $a = 2\pi^2 \cos \pi t$ . At time  $t = 0$ , the object is at the point  $x = 0$  and has a velocity  $v = \pi$  metres per second.
  - (a) Find the velocity  $v$  as a function of  $t$ , for  $t \geq 0$ .
  - (b) Find the values of  $t$ , for  $0 \leq t \leq 4$ , for which the particle is at rest.
  - (c) Show that the displacement,  $x$ , is given by  $x = -2\cos \pi t + \pi t + 3$  for  $t \geq 0$ .
  - (d) Show that the maximum displacement of  $x$  is given by  $x = (3 + \sqrt{3}) + \frac{19\pi}{6}$ , for  $0 \leq t \leq 4$ .
- 8 The velocity  $v$ , in metres per second, of a particle travelling in a straight line is given by  $v = 4 - 2t$  at time  $t$  seconds. The displacement  $x$  of the particle at time  $t = 0$  is given by  $x = 1$ .
  - (i) Find the displacement  $x$  of the particle at  $t = 1$  second.
  - (ii) Find the acceleration of the particle when it is stationary.
  - (iii) Find the distance travelled by the particle in the 4th second.
- 9 A particle  $P$  is initially at the origin and moves so that its velocity is given by  $v = \frac{1}{t+2}$  for  $t \geq 0$ .
  - (i) Find the acceleration of  $P$  when  $t = 2$ .
  - (ii) What is the displacement  $x$  of  $P$  from the origin when  $t = 2$ ?
- 10 A car started from rest with a constant acceleration for the first 10 seconds until it reached a velocity of 25 metres per second. For the next 20 seconds the car maintained a constant velocity of 25 metres per second. At this point in time the brakes were applied. The car slowed down at a constant rate so that 5 seconds later it had a velocity of 5 metres per second.

- (i) Let the car's speed be  $v$  m/s. Draw a graph of  $v$  as a function of time  $t$ , measured in seconds.
- (ii) Let  $s$  metres be the distance travelled by the car from where it had stopped. On a separate set of axes, draw a graph of  $s$  as a function of time and find the distance travelled.

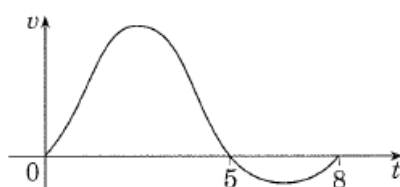
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A particle moves in a straight line. The graph above shows the distance  $x$  of the particle from a fixed point at time  $t$ .

- (i) Is the particle moving faster at time  $t_3$  or  $t_4$ ? Explain.
- (ii) What is the velocity at time  $t_4$ ? Explain.
- (iii) Sketch the graph of the velocity  $v$  as a function of the time  $t$ .

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A graph of velocity  $v$  as a function of time  $t$  is shown above and represents a particle moving in a straight line between  $t = 0$  and  $t = 8$ .

Copy this graph into your Writing Booklet.

- (i) On the  $t$ -axis, clearly mark and label with the letter  $A$  the times when the acceleration of the particle is zero.
- (ii) On the  $t$ -axis, clearly mark and label with the letter  $B$  the time when the acceleration of the particle is largest.
- (iii) The particle is at rest when  $t = 0$ ,  $t = 5$  and  $t = 8$ . Indicate, giving reasons for your answer, when the particle is furthest from its initial position.

- 13  $t$  minutes after a pump starts operating, the rate of water discharge,  $R$  litres per minute, is given by the relation

$$R = 10 + \frac{10}{1+2t}.$$

- (a) Draw a sketch of  $R$  as a function of  $t$ .
- (b) What is the rate of water discharge,  $R$ , after 7 minutes?
- (c) What value does  $R$  approach as  $t$  becomes very large?
- (d) Calculate the total amount of water discharged in the first 7 minutes.

- 14 A cylindrical tank had 200 litres of water in it. More water was pumped into the tank for twenty minutes until it was full. The rate of flow of the water  $R$ , in litres per minute, into the tank was given by  $R = 80 - 4t$ .

- (a) Write down a formula for the volume  $V$  of water in the tank after  $t$  minutes where  $t \leq 20$ .
- (b) What is the capacity of the tank when it is full?
- (c) How long did it take for the tank to be half full?

- 15 The rate of non-recyclable rubbish  $R$ , in tonnes per week, accumulating at a council rubbish dump is given by

$$R = 100 + \left( \frac{40}{1+t} \right)^2,$$

where  $t$  is the time in years.

A new recycling plant at the rubbish dump has been installed to progressively reduce the amount of non-recyclable rubbish over a number of years.

- (i) What is the rate of non-recyclable rubbish  $R$  just prior to the commencement of the recycling plant?
- (ii) What would be the rate of non-recyclable rubbish  $R$  at the start of the tenth year since the commencement of the recycling plant?
- (iii) What value does  $R$  approach over a long period of time?
- (iv) Draw a sketch of  $R$  as a function of  $t$ .
- (v) Calculate the total amount of non-recyclable rubbish deposited at the rubbish dump over the first nine years in which the recycling plant is operational.
- 16 The population  $P(t)$  of ants in a particular colony is given by

$$P(t) = \frac{A}{2 + e^{-t}},$$

where  $A$  is a constant and the time  $t$  (for which  $t \geq 0$ ), is measured in months.

- (i) It is estimated that  $P(t) = 3 \times 10^5$  when  $t = 0$ . Find the value of  $A$ .
- (ii) Calculate the value of  $P(t)$  after 1 month.
- (iii) Estimate, giving reasons, the number of ants expected to be found in the colony over a long period of time.
- (iv) Express the rate at which the number of ants in the colony increases over a period of time,  $t$  months.

- 17 The population  $P(t)$  of persons in a new suburb increases at a rate given by the equation

$$\frac{dP}{dt} = kP,$$

where  $k$  is a constant, and  $t$  is the time in years. The population of the suburb is expected to double every fifteen years.

It may be assumed that  $P = P_0 e^{kt}$  where  $P_0$  is the initial population.

- (a) Find the value of  $k$ .
  - (b) In which year will the suburb attain a population four times that which it had at the beginning of the year 2000?
  - (c) It is known the population of the suburb at the beginning of the year 2000 was 30 000. What is the population expected to be in the year 2035?
- 18 Two regional towns  $X$  and  $Y$  have populations,  $N$ , of the towns given by

$$N_X = 3000e^{-0.02t} \text{ for town } X, \text{ and}$$

$$N_Y = 2000e^{0.03t} \text{ for town } Y,$$

where  $t$  is the number of years which have passed since 1 January 2000.

- (i) Write down the annual growth rate for town  $X$ .
- (ii) Calculate the instantaneous rate at which the population of town  $Y$  will be increasing at 1 January 2015.
- (iii) In which year will the population of town  $Y$  become larger than the population of town  $X$ ?
- (iv) On the same set of axes sketch the graphs of both towns  $X$  and  $Y$  for the period 1 January 2000 to 1 January 2020. Indicate on your graph when the population of  $Y$  exceeds that of  $X$ .

- 19 A quantity  $Q$  of radioactive material at time  $t$  in years is given by  $Q = Q_0 e^{-kt}$ , where  $k$  is a constant and  $Q_0$  is the amount of radioactive material at time  $t = 0$ .

- (i) Given that  $Q = \frac{1}{2}Q_0$ , when  $t = 1500$  years, calculate  $k$ .
  - (ii) After how many years does only 15% of the initial amount of radioactive material remain?
- 20 Ten grams of salt is placed in a container of water and begins to dissolve. After  $t$  minutes the amount  $A$  grams of undissolved salt is given by  $A = A_0 e^{-kt}$ .
- (i) Find the value of  $A_0$ .
  - (ii) Calculate  $k$ , given that  $A = 3.2$  when  $t = 4$ .
  - (iii) How many minutes will elapse for 1 gram of salt to remain undissolved?

- 21 The number  $N$  of dust particles in a room satisfies the equation  $N = N_0 e^{0.15t}$ , where  $t$  is measured in days and  $N_0$  is a constant.

- (i) Show that the number of dust particles increases at a rate proportional to the number present.
- (ii) When  $t = 5$ , the number of dust particles was estimated to be  $1.8 \times 10^9$ . Evaluate  $N_0$ .
- (iii) The number of dust particles doubles every  $x$  days. Find  $x$ .

# APPLICATIONS OF CALCULUS TO THE PHYSICAL WORLD

## Worked solutions to examination questions

1 (a)  $x = t^3 - 6t^2 + 1$

$$v = \frac{dx}{dt} = 3t^2 - 12t$$

$$= 3t(t - 4)$$

= 0 when the particle is stationary

$$\therefore t = 0, t = 4$$

i.e. at rest at  $t = 0$  seconds and  $t = 4$  seconds

(b)  $v = \frac{dx}{dt} = 3t^2 - 12t$

$$a = \frac{d^2x}{dt^2} = 6t - 12 \text{ ms}^{-2}$$

2  $x = 2 - 4 \sin 2t$

(i) At  $t = 0$ ,  $x = 2 - 4 \sin 0 = 2 \text{ m}$

(ii)  $v = \frac{dx}{dt} = -4(2) \cos 2t$

$$= -8 \cos 2t$$

= 0 when the particle is stationary

$$\therefore \cos 2t = 0$$

$$\therefore 2t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$\therefore t = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \dots$$

When  $t = \frac{\pi}{4}$ ,  $x = 2 - 4 \sin 2t$

$$= 2 - 4 \sin 2\left(\frac{\pi}{4}\right)$$

$$= 2 - 4 \sin \frac{\pi}{2}$$

$$= 2 - 4$$

$$= -2 \text{ m}$$

$\therefore$  the particle first comes to rest 2 m to the left of the origin after  $\frac{\pi}{4}$  seconds.

(iii) The particle is next stationary at  $\frac{3\pi}{4}$  seconds (see (ii)).

(iv)  $v = \frac{dx}{dt} = -8 \cos 2t$

$$a = \frac{d^2x}{dt^2} = -8(2)(-\sin 2t) = 16 \sin 2t$$

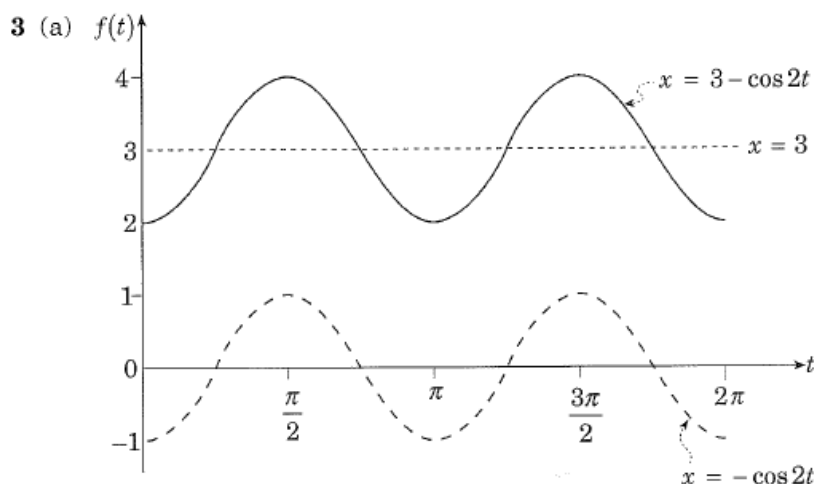
When  $t = \frac{\pi}{12}$ ,  $a = 16 \sin 2\left(\frac{\pi}{12}\right)$

$$= 16 \sin \frac{\pi}{6}$$

$$= 16\left(\frac{1}{2}\right)$$

$$= 8 \text{ ms}^{-2}$$

Factorise



The graph of  $x = 3 - \cos 2t$  is formed by combining the graphs of  $x = 3$  and  $x = -\cos 2t$ .

(b)  $x = 3 - \cos 2t$

$$v = \frac{dx}{dt} = -2(-\sin 2t) = 2\sin 2t$$

= 0 when the particle is stationary

$$\therefore \sin 2t = 0$$

$$2t = 0, \pi, 2\pi, 3\pi, 4\pi \text{ seconds for } 0 \leq 2t \leq 4\pi$$

$$t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi \text{ seconds for } 0 \leq t \leq 2\pi$$

From the graph, we have at  $t = 0$  s,  $x = 2$  m

$$t = \frac{\pi}{2} \text{ s, } x = 4 \text{ m}$$

$$t = \pi \text{ s, } x = 2 \text{ m}$$

$$t = \frac{3\pi}{2} \text{ s, } x = 4 \text{ m}$$

$$t = 2\pi \text{ s, } x = 2 \text{ m}$$

(c) The particle reaches maximum velocity when acceleration  $a = 0$

$$\therefore a = \frac{dv}{dt} = 2(2)\cos 2t = 4\cos 2t = 0$$

$$\therefore \cos 2t = 0$$

$$2t = \frac{\pi}{2}, \frac{3\pi}{2}, \dots \text{ for } 0 \leq 2t \leq 4\pi$$

$$\text{i.e. } t = \frac{\pi}{4}, \frac{3\pi}{4}, \dots$$

$\therefore$  the maximum velocity is first reached when  $t = \frac{\pi}{4}$  seconds.

4  $x = t^3 - 12t$

(i)  $v = \frac{dx}{dt} = 3t^2 - 12 \text{ ms}^{-1}$

(ii) At  $t = 3$ ,  $v = 3(3^2) - 12 = 27 - 12 = 15 \text{ ms}^{-1}$

(iii) The particle is stationary when  $v = 0$

$$\therefore v = 3t^2 - 12 = 0$$

$$\text{i.e. } 3(t^2 - 4) = 0$$

$$3(t-2)(t+2) = 0$$

$$\therefore t = 2, -2$$

Rejecting  $t = -2$ , since  $t \geq 0$

$\therefore t = 2$  seconds when first at rest

5  $x = 5 - t + 4 \ln(t+2)$

(i) At  $t = 0$ ,  $x = 5 - 0 + 4 \ln 2 = 5 + 4 \ln 2$  metres

(ii)  $v = \frac{dx}{dt} = -1 + 4 \left( \frac{1}{t+2} \right) = \left( -1 + \frac{4}{t+2} \right) \text{ms}^{-1}$

$$\begin{aligned} a &= \frac{d^2x}{dt^2} = \frac{d}{dt} \left[ -1 + 4(t+2)^{-1} \right] \\ &= 0 + 4(-1)(t+2)^{-2} \\ &= -4(t+2)^{-2} \\ &= -\frac{4}{(t+2)^2} \text{ms}^{-2} \end{aligned}$$

(iii) The particle is stationary when  $v = 0$

$$\therefore v = -1 + \frac{4}{t+2} = 0$$

$$\therefore \frac{4}{t+2} = 1$$

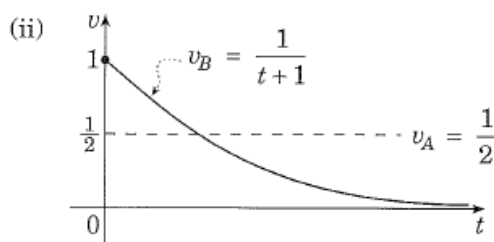
$$t+2 = 4$$

$$t = 2 \text{ s}$$

$\therefore$  The particle is at rest when  $t = 2$  seconds.

6  $x_A = \frac{1}{2}t + 1$ ;  $x_B = \ln(t+1)$

(i)  $v_A = \frac{dx_A}{dt} = \frac{1}{2}$ ;  $v_B = \frac{dx_B}{dt} = \frac{1}{t+1}$



From the graph it can be seen that  $v_A$  and  $v_B$  intersect at only one point and therefore reach the same velocity only once.

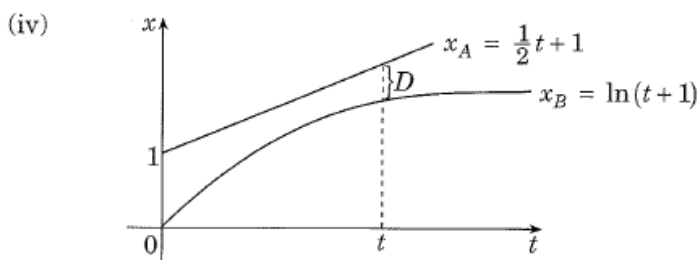
(iii) When  $v_A = v_B$ , then  $\frac{1}{2} = \frac{1}{t+1}$

$$\therefore t+1 = 2$$

$$t = 1 \text{ second}$$

$$\therefore x_B = \ln(1+1) \text{ when } t = 1$$

$$\text{i.e. } x_B = \ln 2$$



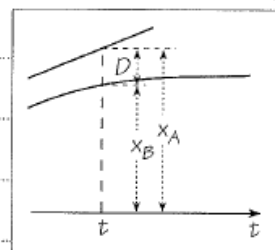
The graphs of  $x_A$ ,  $x_B$  are drawn above. Let  $D$  be the minimum distance between  $A$  and  $B$ . The particles will never meet if the minimum distance is greater than zero, i.e. if  $D > 0$ .

$$D = x_A - x_B = \frac{1}{2}t + 1 - \ln(t+1)$$

$$\therefore \frac{dD}{dt} = \frac{1}{2} - \frac{1}{t+1}$$

$$= 0 \text{ for stationary values}$$

continued ...



$$\therefore \frac{1}{2} = \frac{1}{t+1}$$

$$t+1 = 2$$

$$\therefore t = 1 \text{ second}$$

$$\text{At } t = 1, \quad D = \frac{1}{2}t + 1 - \ln(t+1)$$

$$= \frac{1}{2}(1) + 1 - \ln 2$$

$$= \frac{3}{2} - \ln 2$$

$$\approx 0.81 \text{ cm}$$

$$\frac{d^2D}{dt^2} = \frac{d}{dt} \left[ \frac{1}{2} - (t+1)^{-1} \right]$$

$$= -(-1)(t+1)^{-2}$$

$$= \frac{1}{(t+1)^2} > 0 \text{ for all } t$$

$\therefore$  a minimum occurs for  $D$  at  $t = 1$  second,

i.e. the minimum distance ( $D \approx 0.81$  cm) occurs at  $t = 1$

$\therefore$  the particles  $A$  and  $B$  never meet.

$$7 \quad a = \frac{d^2x}{dt^2} = 2\pi^2 \cos \pi t$$

$$(a) \quad v = \int a \, dt = \int 2\pi^2 \cos \pi t \, dt$$

$$= 2\pi^2 \left( \frac{\sin \pi t}{\pi} \right) + c_1$$

$$= 2\pi \sin \pi t + c_1$$

$$\therefore \pi = 2\pi \sin \pi(0) + c_1 \quad \text{for } v = \pi, t = 0$$

$$= 2\pi \sin 0 + c_1$$

$$\therefore c_1 = \pi$$

$$\therefore v = 2\pi \sin \pi t + \pi$$

(b) The particle is at rest when  $v = 0$ ,

$$\therefore 0 = 2\pi \sin \pi t + \pi$$

$$-\pi = 2\pi \sin \pi t$$

$$\sin \pi t = -\frac{1}{2} \text{ for } 0 \leq t \leq 4, \text{ i.e. } 0 \leq \pi t \leq 4\pi$$

$$\pi t = \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}, 3\pi + \frac{\pi}{6}, 4\pi - \frac{\pi}{6}$$

$$\text{i.e. } \pi t = \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{19\pi}{6}, \frac{23\pi}{6}$$

$$\therefore t = \frac{7}{6}, \frac{11}{6}, \frac{19}{6}, \frac{23}{6} \text{ seconds for } 0 \leq t \leq 4$$

(c) Since  $v = 2\pi \sin \pi t + \pi$

$$x = \int v \, dt = \int (2\pi \sin \pi t + \pi) \, dt$$

$$= 2\pi \left( \frac{-\cos \pi t}{\pi} \right) + \pi t + c_2$$

$$\text{i.e. } x = -2 \cos \pi t + \pi t + c_2$$

$$\text{When } t = 0, \quad x = 1$$

$$\therefore 1 = -2 \cos 0 + 0 + c_2$$

$$= -2 + c_2$$

$$c_2 = 3$$

$$\therefore x = -2 \cos \pi t + \pi t + 3$$

The maximum value of  $x$  must occur at  $v = 0$ ,

i.e. one of the values of  $t = \frac{7}{6}, \frac{11}{6}, \frac{19}{6}, \frac{23}{6}$

$$\text{At } t = \frac{7}{6}, -2\cos\frac{7\pi}{6} > 0$$

$$\text{at } t = \frac{11}{6}, -2\cos\frac{11\pi}{6} < 0$$

$$\text{at } t = \frac{19}{6}, -2\cos\frac{19\pi}{6} > 0$$

$$\text{at } t = \frac{23}{6}, -2\cos\frac{23\pi}{6} < 0$$

$\therefore x$  will be a maximum for either  $t = \frac{7}{6}$  or  $\frac{19}{6}$ .

$$\text{Also for the term } \pi t, \frac{19\pi}{6} > \frac{7\pi}{6}$$

$\therefore x$  is a maximum when  $t = \frac{19}{6}$ .

$$\begin{aligned}\therefore x &= -2\cos\frac{19\pi}{6} + \frac{19\pi}{6} + 3 \\ &= -2\left(-\frac{\sqrt{3}}{2}\right) + \frac{19\pi}{6} + 3 \\ &= \frac{19\pi}{6} + 3 + \sqrt{3} \\ &= (3 + \sqrt{3}) + \frac{19\pi}{6}\end{aligned}$$

$$\begin{aligned}8 \quad x &= \int v \, dt = \int (4 - 2t) \, dt \\ &= 4t - t^2 + c\end{aligned}$$

$$\text{At } t = 0, x = 1$$

$$\therefore 1 = 4(0) - 0 + c$$

$$c = 1$$

$$\therefore x = 4t - t^2 + 1$$

$$(i) \text{ When } t = 1, \quad x = 4(1) - 1^2 + 1 = 4 \text{ m}$$

$$(ii) \quad v = 4 - 2t$$

$$a = \frac{dv}{dt} = -2 \text{ ms}^{-2} \text{ for all } t$$

$$(iii) \text{ Distance travelled in the 4th second}$$

$$= (\text{distance travelled in 4 s}) - (\text{distance travelled in 3 s})$$

$$= \left| \int_0^4 (4 - 2t) \, dt - \int_0^3 (4 - 2t) \, dt \right|$$

$$= \left| [4t - t^2]_0^4 - [4t - t^2]_0^3 \right|$$

$$= |(16 - 16) - 0 - \{4(3) - 9\} - 0|$$

$$= |0 - (3)|$$

$$= 3$$

$\therefore$  the particle travels 3 m in the 4th second.

$$9 \quad v = \frac{1}{t+2} \text{ for } t \geq 0 \text{ and } x = 0 \text{ when } t = 0$$

$$\begin{aligned}(i) \quad a &= \frac{dv}{dt} = \frac{d}{dt}[(t+2)^{-1}] \\ &= -(t+2)^{-2}\end{aligned}$$

The acceleration  $a = -2 \text{ ms}^{-2}$  is constant.

Absolute value is required since for  $2 < t \leq 4$  the particle is travelling in the opposite direction to that when  $0 \leq t < 2$ .

continued ...



$$= -\frac{1}{(t+2)^2}$$

$$\text{When } t = 2, \quad a = -\frac{1}{(2+2)^2} = -\frac{1}{4^2} = -\frac{1}{16} \text{ ms}^{-2}$$

$$(ii) \quad x = \int v \, dt = \int \frac{1}{t+2} \, dt$$

$$x = \ln(t+2) + c$$

$$\text{When } t = 0, \quad x = 0$$

$$\therefore 0 = \ln(0+2) + c$$

$$\therefore c = -\ln 2$$

$$x = \ln(t+2) - \ln 2$$

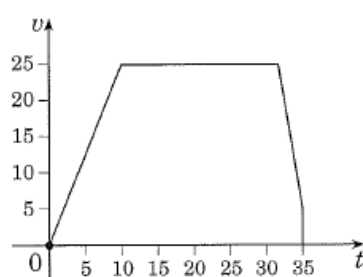
$$\text{When } t = 2, \quad x = \ln(2+2) - \ln 2$$

$$= \ln 4 - \ln 2$$

$$= \ln\left(\frac{4}{2}\right)$$

$$= \ln 2$$

10 (i)



(ii) The shape of the displacement / time curve is determined by the velocity / time relationships.

For  $0 \leq t \leq 10$  seconds, the  $v/t$  relationship is linear,

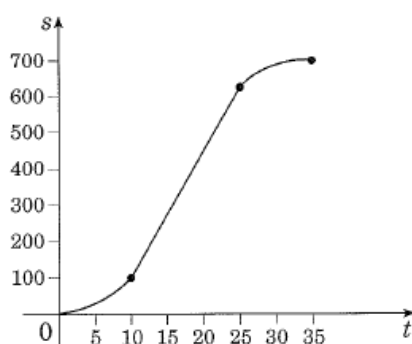
$\therefore$  the  $s/t$  relationship is parabolic.

For  $10 \leq t \leq 30$  seconds, the  $v/t$  relationship is constant,

$\therefore$  the  $s/t$  relationship is linear.

For  $30 \leq t \leq 35$  seconds, the  $v/t$  relationship is linear,

$\therefore$  the  $s/t$  relationship is parabolic.



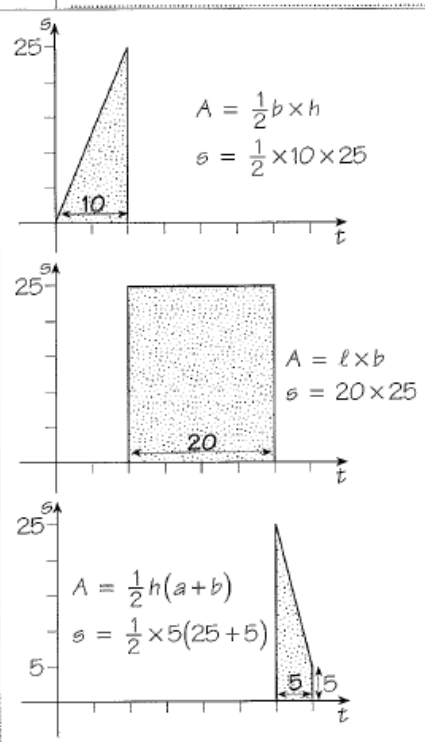
The distance travelled is equivalent to the area under the  $v/t$  graph.

$$\text{For } 0 \leq t \leq 10 \text{ s,} \quad s = \frac{1}{2} \times 10 \times 25 = 125 \text{ m}$$

$$\text{For } 10 \leq t \leq 30 \text{ s,} \quad s = 20 \times 25 = 500 \text{ m}$$

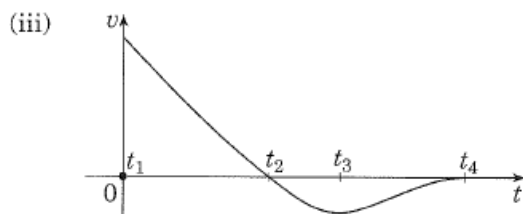
$$\text{For } 30 \leq t \leq 35 \text{ s,} \quad s = \frac{1}{2} \times 5(25+5) = 75 \text{ m}$$

$$\therefore \text{ Total distance travelled} = 700 \text{ m}$$

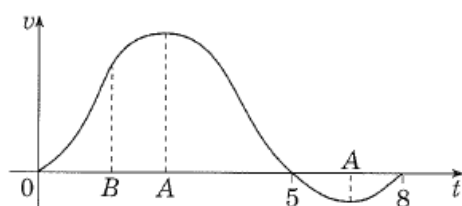


- 11 (i) The absolute value of the slope of the curve is greater at  $t_3$  than at  $t_4$ . Since the slope (gradient) is  $dx/dt$  and represents the velocity, the velocity is greater at  $t_3$  than at  $t_4$ .

- (ii) At  $t = 4$ , the slope appears to be parallel to the  $x$ -axis and therefore to be zero. Therefore the velocity would be zero.



- 12 (i), (ii)



- (iii) For  $0 \leq t \leq 5$ , the velocity is positive, therefore the particle is moving away from the initial starting position.

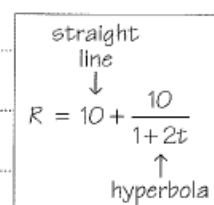
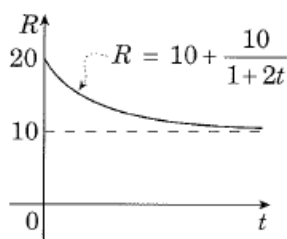
For  $5 \leq t \leq 8$ , the velocity is negative, therefore the particle is moving back towards the starting position.

Thus the particle is furthest from the initial position at  $t = 5$ .

- 13 (a) The relationship  $R = 10 + \frac{10}{1+2t}$  is made up of the addition of the curves  $R = 10$  (a straight line parallel to the  $x$ -axis) and  $R = \frac{10}{1+2t}$  (a hyperbola in the first quadrant).

$$\begin{aligned} \text{At } t = 0, \quad R &= 10 + \frac{10}{1+0} \\ &= 10 + 10 \\ &= 20 \end{aligned}$$

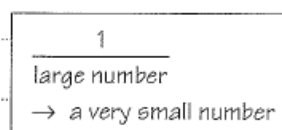
$$\begin{aligned} \text{As } t \rightarrow \infty, \quad R &\rightarrow 10 + \frac{10}{\infty} \\ &\rightarrow 10 + 0 \\ &\rightarrow 10 \end{aligned}$$



$$\begin{aligned} \text{(b) At } t = 7, \quad R &= 10 + \frac{10}{1+2(7)} \\ &= 10 + \frac{10}{15} \\ &= 10\frac{2}{3} \text{ L/min} \end{aligned}$$

$$\begin{aligned} \text{(c) As } t \rightarrow \infty, \quad R &\rightarrow 10 + \frac{10}{\infty} \\ &\rightarrow 10 + 0 \\ &\rightarrow 10 \text{ L/min} \end{aligned}$$

i.e. as  $t$  increases,  $R \rightarrow 10 \text{ L/min}$



- (d) Let  $L$  be the volume discharged (in litres).

$$\begin{aligned}\frac{dL}{dt} &= R \\ \therefore L &= \int_0^7 R \, dt \\ &= \int_0^7 \left(10 + \frac{10}{1+2t}\right) dt \\ &= \left[10t + 10\left(\frac{1}{2}\right)\ln(1+2t)\right]_0^7 \\ &= \left[10t + 5\ln(1+2t)\right]_0^7 \\ &= 70 + 5\ln 15 - 0 \\ &= (70 + 5\ln 15) \text{ litres}\end{aligned}$$

- 14 (a) The flow rate ( $R$ ) of the volume ( $V$ ) is given by  $\frac{dV}{dt} = R$

$$\therefore V = \int R \, dt = \int (80 - 4t) \, dt = 80t - 2t^2 + c$$

$$\text{At } t = 0, \quad V = 200$$

$$\therefore 200 = 80(0) - 2(0) + c$$

$$c = 200$$

$$\therefore V = 80t - 2t^2 + 200$$

- (b) The tank is full after 20 minutes.

$$\begin{aligned}\text{At } t = 20, \quad V &= 80(20) - 2(20^2) + 200 \\ &= 1600 - 800 + 200 \\ &= 1000 \text{ litres}\end{aligned}$$

- (c) The tank is half full when  $V = 500$  litres

$$\therefore 500 = 80t - 2t^2 + 200$$

$$2t^2 - 80t + 300 = 0$$

$$t^2 - 40t + 150 = 0$$

$$\begin{aligned}\therefore t &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-40) \pm \sqrt{(-40)^2 - 4(1)(150)}}{2(1)} \\ &= \frac{40 \pm \sqrt{1600 - 600}}{2} \\ &= \frac{40 \pm \sqrt{1000}}{2} \\ &= \frac{40 \pm 10\sqrt{10}}{2} \\ &= 20 \pm 5\sqrt{10} \\ &= 35.8 \text{ and } 4.2 \text{ minutes}\end{aligned}$$

Since  $t \leq 20$ , then  $t = 35.8$  min is rejected.

$\therefore$  The time to half fill the tank is given by

$$t = (20 - 5\sqrt{10}) \text{ minutes } (\approx 4.2 \text{ min}).$$

$$15 \quad R = 100 + \left(\frac{40}{1+t}\right)^2$$

- (i) Initially  $t = 0$ ,

$$\therefore R = 100 + \left(\frac{40}{1+0}\right)^2 = 100 + 1600 = 1700 \text{ tonnes / year}$$

$R$  is the rate at which the discharge  $L$  is occurring over a period of time  $t$ ,

$$\text{i.e.} \quad R = \frac{dL}{dt}$$

$$\int \frac{10}{1+2t} \, dt = 5 \int \frac{2}{1+2t} \, dt$$

$$\Rightarrow \int \frac{f'(t)}{f(t)} \, dt \Rightarrow \log f(t)$$

(ii) The start of the tenth year  $\Rightarrow t = 9$  years

$$\therefore R = 100 + \left(\frac{40}{1+9}\right)^2 = 100 + (4)^2 = 116 \text{ tonnes/year}$$

(iii) As  $t \rightarrow \infty$ ,  $(1+t) \rightarrow \infty$

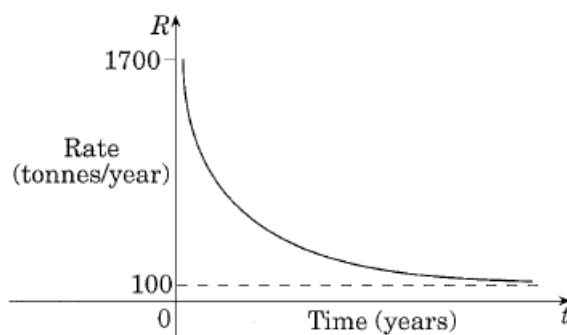
$$\frac{1}{1+t} \rightarrow 0$$

$$\frac{40}{1+t} \rightarrow 0$$

$$\therefore R = 100 + \left(\frac{40}{1+t}\right)^2 \rightarrow 100 + 0^2$$

i.e.  $R \rightarrow 100$  tonnes/year

(iv)



(v) Let  $M$  be the total amount of non-recyclable rubbish.

$$\therefore \frac{dM}{dt} = R$$

$$\begin{aligned} \therefore M &= \int R \, dt = \int_0^9 100 + \left(\frac{40}{1+t}\right)^2 dt \\ &= \int_0^9 100 + 1600(1+t)^{-2} dt \\ &= \left[ 100t + 1600 \left( \frac{1+t}{-1} \right)^{-1} \right]_0^9 \\ &= \left[ 100t - \frac{1600}{1+t} \right]_0^9 \\ &= \left[ 100(9) - \frac{1600}{10} \right] - \left[ 100(0) - \frac{1600}{1} \right] \\ &= 900 - 160 + 1600 \\ &= 2340 \text{ tonnes} \end{aligned}$$

Total amount  $\Rightarrow$  some mass (say  $M$ )

$$\frac{dM}{dt} = R \text{ (the rate of rubbish)}$$

What process enables you to find  $M$ ?

**16**  $P(t) = \frac{A}{2+e^{-t}}$

(i) At  $t = 0$ ,  $P(t) = 3 \times 10^5$

$$3 \times 10^5 = \frac{A}{2+e^0} = \frac{A}{3} \text{ since } e^0 = 1$$

$$\therefore A = 9 \times 10^5$$

(ii) At  $t = 1$ ,  $P(1) = \frac{9 \times 10^5}{2+e^{-1}}$

$$= 380\,086.9 \dots$$

$$\approx 380\,000 \text{ to the nearest thousand}$$

(iii) As  $t \rightarrow \infty$ ,  $e^{-t} \rightarrow \frac{1}{e^\infty} \rightarrow 0$

$$\therefore P(t) \rightarrow \frac{9 \times 10^5}{2+0}$$

$$\text{i.e. } P(t) \rightarrow 4.5 \times 10^5$$

(iv) The rate of increase of the ant population is given by  $\frac{dP}{dt}$ .

$$\begin{aligned} P &= \frac{A}{2+e^{-t}} = A(2+e^{-t})^{-1} \\ \frac{dP}{dt} &= A(-1)(2+e^{-t})^{-2}(-e^{-t}) \\ &= \frac{Ae^{-t}}{(2+e^{-t})^2} \\ &= \frac{A}{e^t(2+e^{-t})^2} \\ &= \frac{9 \times 10^5}{e^t(2+e^{-t})^2} \text{ ants / month} \end{aligned}$$

$$17 \ P = P_0 e^{kt}$$

(a) At  $t = 15$ ,  $P = 2P_0$

$$\therefore 2P_0 = P_0 e^{k(15)}$$

$$\therefore 2 = e^{15k}$$

$$15k = \ln 2$$

$$k = \frac{\ln 2}{15}$$

(b) Let  $P_0$  be the population in the year 2000

$$\therefore 4P_0 = P_0 e^{kt}$$

$$\therefore 4 = e^{kt}$$

$$kt = \ln 4$$

$$\begin{aligned} \therefore t \left( \frac{\ln 2}{15} \right) &= \ln 4 \quad \text{using } k = \frac{\ln 2}{15} \\ t &= \frac{15 \ln 4}{\ln 2} \\ &= 30 \text{ years} \end{aligned}$$

$\therefore$  the population will have quadrupled by the year 2030.

(c) The year 2035  $\Rightarrow t = 35$  years, and  $P_0 = 30\,000$

$$\begin{aligned} P &= P_0 e^{kt} \\ &= 30\,000 e^{35 \left( \frac{\ln 2}{15} \right)} \\ &= 151\,190 \end{aligned}$$

18 (i) The annual growth rate is given by the proportion constant  $k$ .

$$\text{For town X: } N_X = 3000e^{-0.02t} = 3000e^{kt}$$

$$\therefore k = -0.02$$

(ii) The instantaneous rate of change is given by  $\frac{dN_Y}{dt}$ .

$$\begin{aligned} N_Y &= 2000e^{0.03t} \\ \frac{dN_Y}{dt} &= 2000(0.03)e^{0.03t} \\ &= 60e^{0.03t} \\ &= 60e^{0.03(15)}, \text{ when } t = 15 \text{ years} \\ &= 94.09 \dots \\ &\approx 94 \text{ persons / year} \end{aligned}$$

Chain rule

Double the population  $\Rightarrow P = 2P_0$

$$\frac{\ln 4}{\ln 2} = \frac{\ln 2^2}{\ln 2} = \frac{2 \ln 2}{\ln 2} = 2$$

Since  $k < 0$  for town Y,  
the population is declining.

The year 2015  $\Rightarrow t = 15$  years

- (iii) The population of town Y exceeds the population of town X when  $N_Y > N_X$

i.e.  $2000e^{0.03t} > 3000e^{-0.02t}$

$$2000e^{0.03t} > \frac{3000}{e^{0.02t}}$$

$\therefore 2000e^{0.03t} \times e^{0.02t} > 3000$

$$e^{0.03t + 0.02t} > \frac{3000}{2000}$$

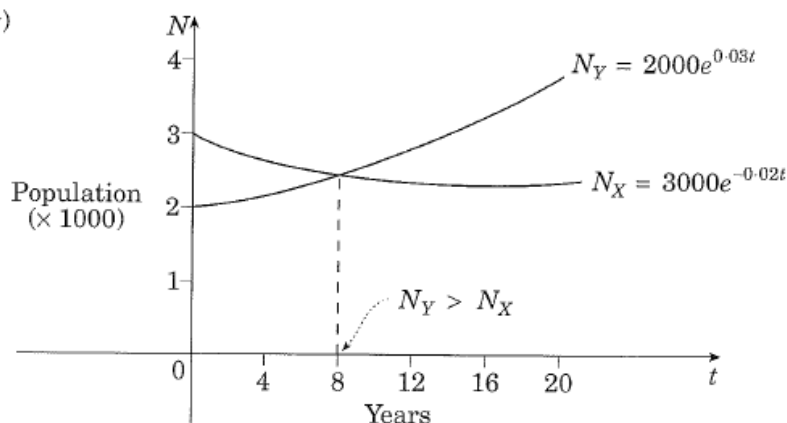
$$e^{0.05t} > 1.5$$

$$0.05t > \ln 1.5$$

$$t > \frac{\ln 1.5}{0.05} = 8.1 \text{ years}$$

$\therefore$  The population of town X will be greater in 2008.

(iv)



19 (i)  $Q = Q_0 e^{-kt}$  At  $t = 1500$ ,  $Q = \frac{1}{2}Q_0$

$$\frac{1}{2}Q_0 = Q_0 e^{-1500k}$$

$$\frac{1}{2} = e^{-1500k}$$

$$-1500k = \ln\left(\frac{1}{2}\right)$$

$$\therefore k = \frac{\ln\left(\frac{1}{2}\right)}{-1500} = 4.62 \times 10^{-4} \text{ (to 2 dec. pl.)}$$

- (ii) After  $t$  years only 15% of  $P_0$ , i.e.  $0.15P_0$  remains

$$\therefore 0.15P_0 = P_0 e^{-kt}$$

$$e^{-kt} = 0.15$$

$$-kt = \ln(0.15)$$

$$t = \frac{\ln(0.15)}{k}$$

$$= \frac{\ln(0.15)}{-4.62 \times 10^{-4}}$$

$$= 4106.32 \dots$$

$$\approx 4106 \text{ years (to the nearest year)}$$

Multiply both sides by  $e^{0.02t}$

Divide both sides by 2000.

When multiplying the same base, add the indices.

**20**  $A = A_0 e^{-kt}$

(i)  $A_0$  is the initial amount when  $t = 0$ ,

$$\therefore A_0 = 10 \text{ g}$$

(ii)  $A = A_0 e^{-kt}$

$$\therefore 3.2 = 10e^{-4k} \text{ when } t = 4$$

$$e^{-4k} = \frac{3.2}{10} = 0.32$$

$$-4k = \ln(0.32)$$

$$\therefore k = \frac{\ln(0.32)}{-4} = 0.2848 \dots$$

(iii)  $A = A_0 e^{-kt}$

$$\therefore 1 = 10e^{-kt}$$

$$e^{-kt} = \frac{1}{10} = 0.1$$

$$-kt = \ln(0.1)$$

$$\begin{aligned} \therefore t &= \frac{\ln(0.1)}{-k} \\ &= \frac{\ln(0.1)}{-0.2848 \dots} \\ &= 8.08 \\ &\approx 8.1 \text{ minutes (to 1 dec. pl.)} \end{aligned}$$

**21** (i)  $N = N_0 e^{0.15t}$

The rate of increase is given by  $\frac{dN}{dt}$ ,

$$\begin{aligned} \therefore \frac{dN}{dt} &= N_0 \frac{d}{dt}(e^{0.15t}) \\ &= N_0 (0.15e^{0.15t}) \\ &= 0.15N_0 e^{0.15t} \\ &= 0.15N \text{ since } N = N_0 e^{0.15t} \end{aligned}$$

i.e.  $\frac{dN}{dt} \propto N$

$\propto$  means 'is proportional to'

(ii)  $N = N_0 e^{0.15t}$

$$1.8 \times 10^9 = N_0 e^{(0.15)5}$$

$$\begin{aligned} \therefore N_0 &= \frac{1.8 \times 10^9}{e^{0.75}} \\ &= 8.5 \times 10^8 \end{aligned}$$

(iii) When  $t = x$ ,  $N = 2N_0$

$$\therefore 2N_0 = N_0 e^{0.15x}$$

$$e^{0.15x} = 2$$

$$0.15x = \ln 2$$

$$= \frac{\ln 2}{0.15}$$

$$= 4.62 \dots$$

$$\approx 4.6 \text{ days (to 1 dec. pl.)}$$